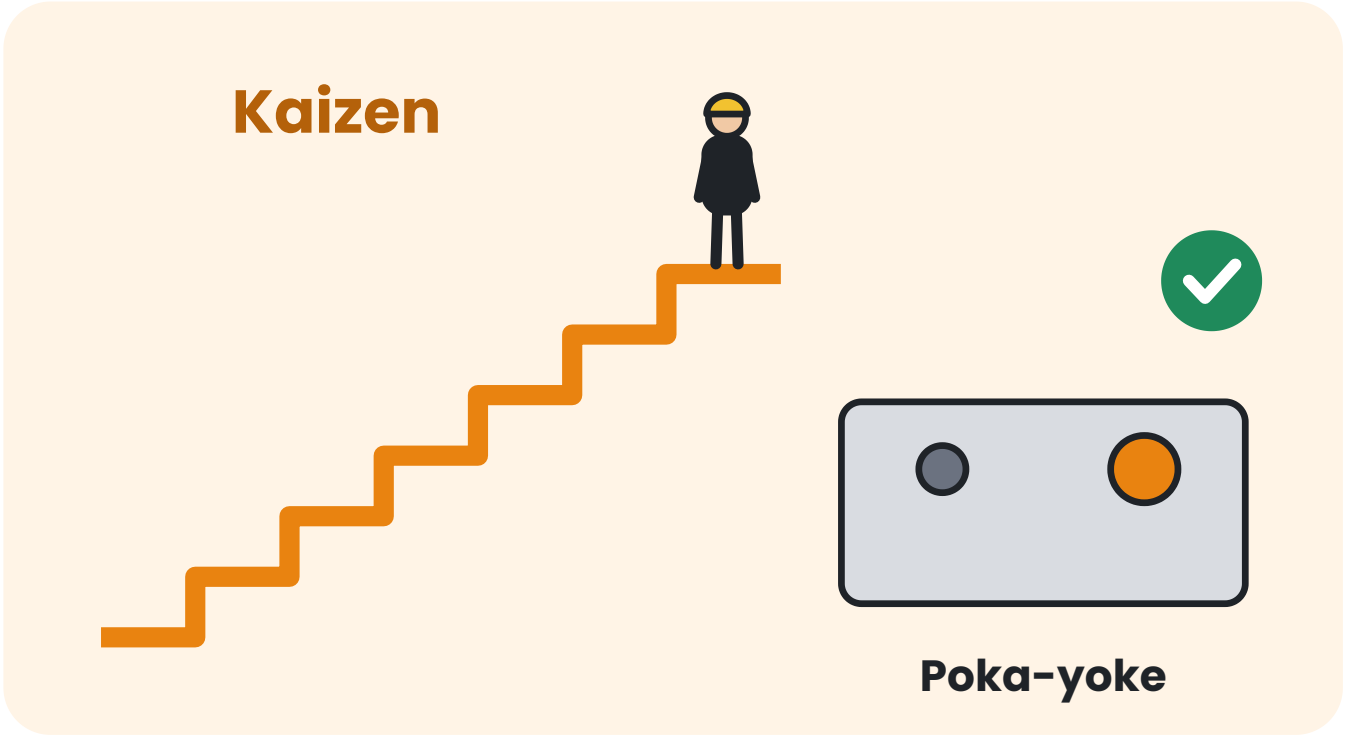
MachiNest

LEAN MANUFACTURING BASICS

Kaizen & Poka-yoke: Improve Every Day, Stop Mistakes

Small daily improvements and simple mistake-proofing for CNC and machine shops, with examples and a printable Kaizen sheet.

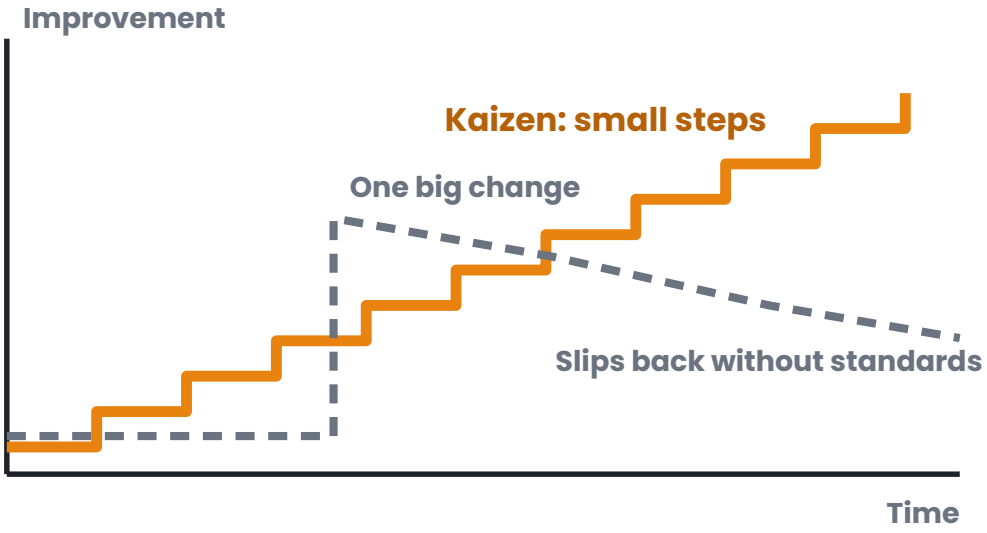
Kaizen | PDCA | 5 Why | Poka-yoke | Kaizen sheet



PART 1 · KAIZEN

What is Kaizen?

Kaizen is a Japanese word: **Kai** means change and **Zen** means good. It means continuous improvement through many small changes, made by everyone, every day.



Many small improvements, each one made standard, add up and stay. One big change without standards often slips back.

Kaizen principles

1. Small changes, made often, are better than waiting for one big project.
2. Go and see the real place (gemba), the real part and the real data.
3. Fix the process, not the person.
4. Use low-cost ideas first: change the method before buying equipment.
5. Every improvement becomes the new standard.
6. Everyone takes part: operators, setters, inspectors and managers.

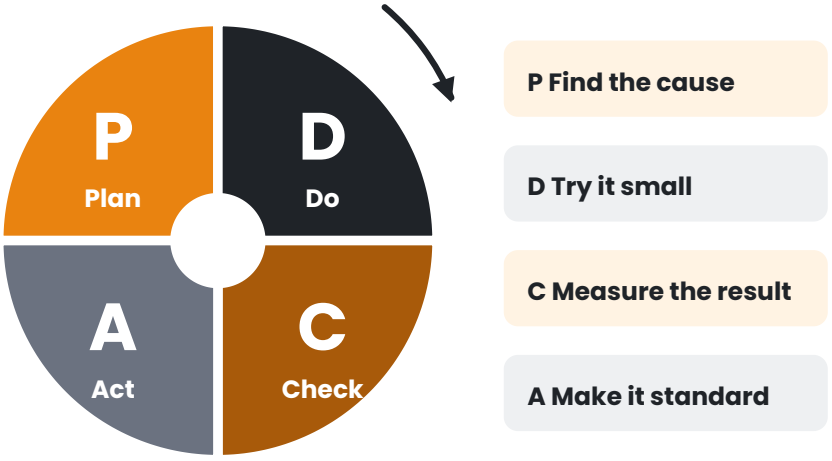
Where to find kaizen ideas

- Repeat rejections and rework
- Long setups and machine stops
- Walking, searching and waiting
- Operator complaints: "this is difficult"
- Safety near misses

PART 1 · KAIZEN

How to run a kaizen: PDCA

PDCA (Plan, Do, Check, Act) is the cycle behind every kaizen. Go round it again and again; each turn makes the process a little better.



Kaizen in 7 steps

1. **Pick one problem** with data: rejects, setup time or downtime.
2. **Go and see** at the machine. Take photos and measure.
3. **Find the root cause** with "5 Why", not the first guess.
4. **Try a small change** on one machine or one shift.
5. **Check the result** against the "before" data.
6. **Make it standard:** update the setup sheet, checklist or program.
7. **Share it** with other shifts and machines, and thank the team.

5 Why example

Problem: bore on part X is oversize on some parts.

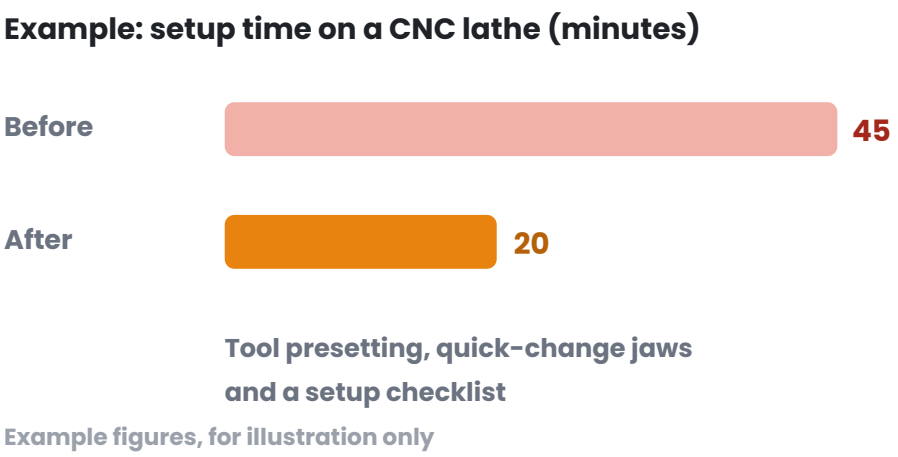
Why 1	The boring bar deflects during the cut.
Why 2	Bar overhang is longer than needed.
Why 3	The shorter bar was not used in this setup.
Why 4	The setup sheet does not list the bar or its overhang.
Why 5	The setup sheet was not updated when the fixture changed.

Action: add the boring bar and its maximum overhang to the setup sheet, and check setup sheets whenever a fixture changes.

PART 1 · KAIZEN

Kaizen examples from a CNC shop

Real kaizens are usually simple. Here are typical ones from machine shops.



Problem	Small change	Result
Long setup on a CNC lathe	Preset tools offline, quick-change soft jaws, setup checklist	Setup time cut by more than half (see chart)
Machine stopped to clear chips from the chuck	Coolant nozzle aimed at the chuck, chip guard added	No stops for chip clearing
Waiting for first-off approval	Trained operators check with go/no-go gauges; QC audits	Less waiting after each setup
Searching for inserts	Labelled insert box at each machine, by code and grade	Inserts found in seconds
Wrong revision drawing used	One drawing holder per machine; old copies removed	No more revision mix-ups

Remember Write down the "before" first: time, count or photo. Without "before" data, no one can see the improvement.

PRINTABLE FORM

Kaizen sheet

Kaizen no. _____ Date _____ Area / machine _____ Team _____

Problem (before)

Photo or sketch

Root cause (5 Why)

Change made (after)

Photo or sketch

Result

	Before	After
Time (setup / cycle)		
Quality (rejects)		
Cost		
Safety / ease		

Made standard? (setup sheet, checklist, program updated)

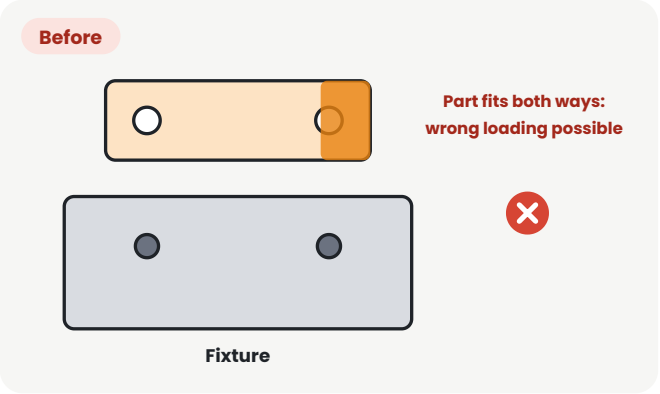
☐ Yes ☐ No

Approved by _____

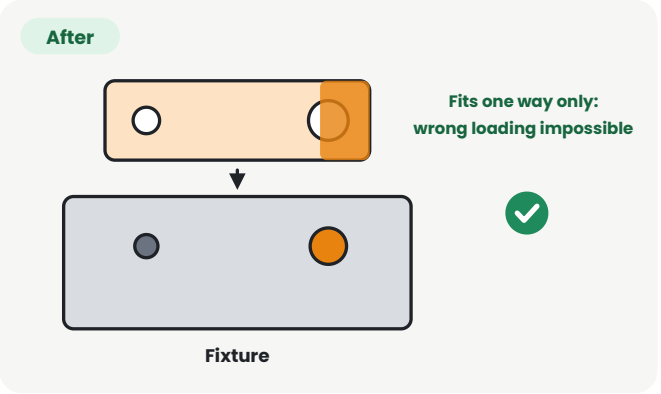
PART 2 · POKA-YOKE

What is Poka-yoke?

Poka-yoke means mistake-proofing. **Poka** means an accidental mistake and **yoke** means to avoid. A poka-yoke is a simple device or method that makes a mistake impossible, or makes it visible at once.



Pins are the same size, so the part can be loaded the wrong way round.



One pin is bigger. The part only fits the right way, so the mistake cannot happen.

Two levels

Prevention (control)

Stops the mistake before it happens, or stops the machine. Best choice.

Detection (warning)

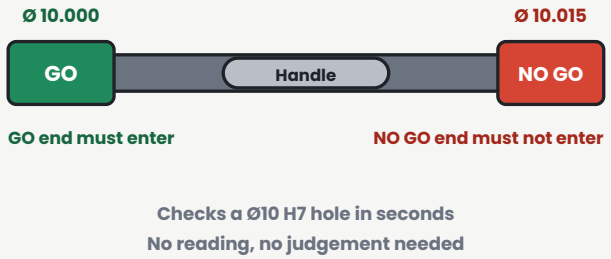
Alerts with a light, sound or message when a mistake happens.

Three methods

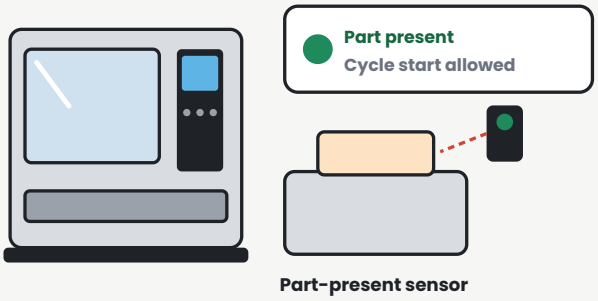
Method	How it works	Shop example
Contact	Checks shape, size or position by touch or sensor	Asymmetric locating pin on a fixture
Fixed value	Checks that a set number of actions or parts is reached	Tool-life counter, part counter at order quantity
Motion step	Checks that steps are done in the right order	Cycle start only after door closed and part clamped

PART 2 · POKA-YOKE

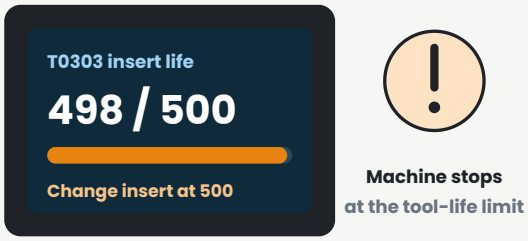
Poka-yoke examples at the machine



A go/no-go plug gauge gives a clear yes/no answer, so no reading or judgement is needed.



A sensor confirms the part is seated before the machine can start.



The control counts parts per insert and stops before the edge fails.

More ideas for CNC shops

- Spindle probe measures a critical size in-cycle and stops if it is out of tolerance
- Broken-tool detection with a tool setter after drilling or tapping
- Different fixtures painted in different colours, matched to a colour on the job card
- Part counter in the program (M-code or macro) stops at the order quantity
- Program simulation and dry run required before the first part
- Door interlock: the spindle cannot run with the door open

PUTTING IT TOGETHER

From problem to poka-yoke

Defect or mistake found → **Find the root cause (5 Why)** → **Kaizen: change the process** → **Poka-yoke: stop it happening again** → **Update the standard**

Questions to find poka-yoke ideas

1. Can this part be loaded the wrong way?
2. Can a wrong tool, offset or program be selected?
3. Can a step be skipped or done out of order?
4. Can a wrong revision or wrong material be used?
5. Can a bad part be mixed with good parts?

Rules of thumb

- Prefer prevention over warning.
- Keep it simple and cheap: a pin, a stop, a colour, a counter.
- Test it: try to make the mistake on purpose and see if it is caught.
- Check poka-yoke devices every shift, like any gauge.

Poka-yoke idea log

Date	Machine / part	Mistake	Poka-yoke idea	Owner	Done